

Defense Answer Sketches: Remaining Questions

Pandemic Trilemma & The Wrong Side of Disruption

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Companion to `defense_answers.tex` (Trilemma framing; Nickell bias; off-support extrapolation). Each question receives a core defense, the necessary check or correction, and existing evidence to invoke.

1 Framing, Scope, and Contribution

1.2 Property 4 redundancy: trilemma or dilemma with shadow constraint?

Defense. The shadow-constraint reformulation is mathematically equivalent but loses two pieces of structure. First, debt is not exogenously bounded: countries face heterogeneous borrowing capacity that is itself state-dependent (sovereign-risk premia rise non-linearly with b/y). Second, the shadow-price interpretation requires the multiplier on the debt constraint to be either taken as given or estimated; the trilemma framing makes the trade-off explicit and observable through realized b .

Compromise. Reframe debt in the introduction as “a co-equal objective with a state-dependent fiscal cost,” acknowledging the formal equivalence to a shadow constraint while preserving the three-objective accounting that the data require.

Existing result. Cross-country variation in cumulative debt accumulation (-2.21 to $+20.64$ pp of 2019 GDP) reveals that countries did not behave as if facing a uniform debt ceiling. The dispersion is the empirical content that distinguishes the trilemma from a constrained dilemma.

1.3 Horizon truncation at $N = 8$ (Q4 2021)

Defense. The cutoff is selected on the marginal condition that the IFR-vaccination interaction crosses zero, not on calendar date. By Q4 2021, $\geq 60\%$ of the OECD population had received at least one dose, and the variant-adjusted IFR had fallen by approximately a factor of three from the pre-vaccine level. This makes Property 1 (containment as the only instrument) no longer binding.

Sensitivity check. Re-estimate Equations (6) and (7) on three windows: Q1 2020–Q4 2021 ($T = 8$), Q1 2020–Q2 2022 (current, $T = 10$), Q1 2020–Q4 2022 ($T = 12$). Report $\hat{\eta}_p$ across windows. The current sample already extends two quarters beyond the binding-constraint horizon to accommodate the lag structure; further extension is not coherent with the structural argument.

Pareto frontier sensitivity. Re-solve the iLQR for each window. If frontiers are stable in shape and rank, the truncation is innocuous. If extending shifts the optimum, this is itself an interesting result and should be reported.

1.4 Why fix S ?

Defense. Three reasons. First, the optimal-containment problem requires epidemiological expertise (variant transmissibility, age-structure-adjusted IFR, behavioral response) that lies outside the economist’s comparative advantage; conditioning on observed S delegates this margin to the public-health literature. Second, observed S varied substantially within countries over time but only narrowly across OECD-average means (range 39–48), making the cross-country fiscal margin the empirically active one. Third, the policy question of interest — compositional optimality of fiscal support given the public-health response — is precisely the second-best problem this design solves.

Quantification of the second-best loss. The full first-best (jointly optimizing S and F) can be computed once the epidemiological calibration is fixed. Report the welfare gap between first-best and conditional-on- S -best. If small, the conditional optimum approximates the unconstrained one; if large, this is a legitimate caveat to acknowledge.

1.5 Scope: split the paper?

Defense. Splitting destroys the chain of identification–calibration–optimization that justifies the framework. The reduced-form panel results are not interesting on their own (no novel estimator, modest sample); the optimization is not credible without the panel-estimated structural parameters; the database is descriptive without the model. The four contributions are jointly necessary.

Concession. An online appendix can carry (i) the codebook for the 2,301-measure database and (ii) the full robustness suite for $\hat{\eta}_p$, leaving the main paper at ~ 50 pages.

2 Theoretical Framework

2.1 Two-instrument mapping: dichotomy or label?

Defense. The dichotomy is operationalized at the measure level: classification follows the economic transmission mechanism (level vs. persistence), not the budgetary category. Above-the-line CP measures (e.g., short-time work) are persistence instruments because they preserve firm-worker matches; they are coded as CP regardless of their statistical treatment. The empirical content is the differential lag structure: CP operates at $k-2$ through interaction with y_{k-1} , while DI operates additively at $k-2$.

Falsification. Re-estimate with above-the-line CP reclassified as DI. If the persistence channel survives only for loans and guarantees, the dichotomy is robust to the contested boundary. Existing memory: PolicyCode 43 (guarantees) drives the entire CP output effect, supporting this falsification.

2.2 Multiplicative vs. additive persistence

Defense. The multiplicative form is structural: capacity preservation modifies the rate of mean-reversion, not the level. A purely additive specification ($\eta_p^{\text{add}} F_{k-2}^{CP}$) is a level injection at $k-2$, observationally similar to DI. The interaction $\eta_p F_{k-2}^{CP} y_{k-1}$ predicts that CP has zero effect when $y_{k-1} = 0$ — the structural prediction that distinguishes CP from DI.

Nesting test. Estimate the augmented specification

$$y_{ik} = \rho_y y_{i,k-1} + \eta_p F_{i,k-2}^{CP} y_{i,k-1} + \eta_p^{\text{add}} F_{i,k-2}^{CP} + \alpha_S S_{ik} + \alpha_{DI} F_{i,k-2}^{DI} + \beta_d d_{ik} + \gamma_i + \delta_k + \varepsilon_{ik}.$$

If η_p^{add} is indistinguishable from zero and η_p retains significance, the multiplicative structure is preferred. The low-stringency null on $\hat{\eta}_p$ already supports this: when $y_{k-1} \approx 0$, the persistence channel vanishes, as the multiplicative form predicts.

2.3 Linear transition system

Defense. Linearity is a deliberate cost-of-tractability choice that the iLQR algorithm requires. Two safeguards: (i) the multiplicative interactions $S \cdot y$ and $F^{CP} \cdot y$ introduce non-linearity in the state-control product; (ii) state and control bounds in iLQR keep the trajectory away from regions where higher-order effects (healthcare-capacity thresholds, hysteresis in firm exit) would dominate.

Sensitivity. Re-estimate Equation (6) with second-order terms in F^{CP} and F^{DI} . If quadratic coefficients are individually and jointly insignificant, the linear approximation is locally valid within the observed support.

2.4 Closure of θ : fear in y but not in θ

Defense. The behavioral response $\beta_d d_k$ in the output equation captures the voluntary withdrawal from contact-intensive activity in response to observed lethality. The same withdrawal also reduces transmission, but the reduced-form θ equation absorbs this through the time-varying effective ρ_θ . A separate fear term in θ would double-count.

Robustness. Add $\beta_\theta d_k$ to Equation (2) and report whether $\hat{\beta}_d$ in the output equation changes. If unchanged, the asymmetric closure is innocuous; if attenuated, the omission biases $\hat{\beta}_d$ upward.

3 Identification: Output Equation

3.2 Effective rank for three-way interactions

Defense. The interactions ψ and η_p are identified from the within-country, within-quarter covariance of $S \cdot y_{k-1}$ and $F^{CP} \cdot y_{k-1}$ with the demeaned outcome. After absorbing 38 country and 10 quarter fixed effects, the residual variation has $380 - 38 - 10 + 1 = 333$ degrees of freedom; the 6 interaction and main-effect coefficients leave 327. The effective sample is large in the regression sense; the more substantive concern is variation, not rank.

Diagnostic. Report the within- R^2 of $S \cdot y_{k-1}$ and $F^{CP} \cdot y_{k-1}$ when regressed on the other regressors and fixed effects. Within- R^2 above 0.9 flags collinearity; below 0.5 confirms identifying variation. Existing memory indicates the within-country correlation $r(S, F^{CP}) = -0.014$ is small, supporting separate identification.

3.3 Fiscal flow: announcement, legislation, or disbursement?

Defense. The 2,301-measure database records the date of formal adoption (legislation or executive decree) and the cost as committed at adoption. This is closer to disbursement timing than announcement timing for transfers and tax measures, which are typically implemented within one quarter of legislation. For loans and guarantees, commitment and disbursement may diverge; the 35% take-up adjustment is an attempt to reconcile.

Robustness. Where IMF or national authorities publish disbursement data (Italy, Germany, France for major schemes), re-estimate using disbursement-timed F^{CP} and F^{DI} . Report whether the lag structure changes.

3.5 Reverse causality on F^{CP}

Defense. The country fixed effect absorbs time-invariant expectations of recovery. The quarter fixed effect absorbs the global pandemic cycle. The residual concern is country-specific time-varying expected growth conditional on which fiscal policy was set — a rational-policy-setting bias.

Test. Project pre-pandemic real-time GDP forecasts (IMF WEO, OECD Economic Outlook, Consensus Economics) on the realized F^{CP} deployment, conditional on γ_i and δ_k . If forecasts predict F^{CP} , this is the channel of bias and the magnitude can be estimated. Existing memory: pairs cluster bootstrap structure is in place; this projection is a one-line addition.

Direction of bias. Countries with pessimistic recovery forecasts likely deployed more F^{CP} . The bias on $\hat{\eta}_p$ is then attenuating, not amplifying: the true effect is more negative than estimated.

3.6 Conditional orthogonality: what could violate it?

Defense. The empirical content of the orthogonality assumption is that, conditional on γ_i and δ_k , fiscal composition is uncorrelated with the unobserved component of output. Plausible threats:

- Sectoral composition shifts within the pandemic (e.g., differential service-sector recovery): partially controlled by including sectoral employment shares from Eurostat as time-varying controls.
- Pre-pandemic banking-sector health: countries with weak banks may have leaned on guarantees more heavily and recovered slower for unrelated reasons. Control with bank-CDS spreads or NPL ratios as of 2019.
- Political-cycle effects: governments facing elections may have over-deployed CP. Control with electoral-calendar dummies (Database of Political Institutions).

Test. Add each control sequentially; report whether $\hat{\eta}_p$ moves. Stability across controls supports the conditional orthogonality.

3.7 Fragility of $\hat{\eta}_p$: distinguishing “no recession” from “no effect”

Defense. The structural prediction is that η_p is non-zero, but the interaction $\eta_p F_{k-2}^{CP} y_{k-1}$ vanishes when either $F^{CP} = 0$ or $y_{k-1} = 0$. The low-stringency subsample has $|y_{k-1}|$ small but F_{k-2}^{CP} also small (low containment $\Rightarrow$ low fiscal need). The product $F^{CP} \cdot y$ is therefore tiny, and the interaction is unidentified.

Test that distinguishes the two hypotheses. Restrict the sample to high- F^{CP} but low-stringency country-quarters (deployment without binding containment, e.g., Sweden). If $\hat{\eta}_p$ remains non-zero in this subsample, the mechanism is general; if zero, the persistence channel operates only through the lockdown-induced damage. Both are interpretable, but the first is the stronger structural claim.

3.8 Functional form of fear

Defense. A linear specification understates the response at high mortality (saturation in attention) and overstates at low mortality (announcement effects). The literature (Atkeson; Eichenbaum–Rebelo–Trabandt) typically uses $\log(1 + d)$ or sigmoid forms.

Robustness. Re-estimate with $\log(1 + d_{ik})$; report whether $\hat{\beta}_d$ remains negative and significant. Existing memory: Goolsbee & Syverson (2021) document that voluntary behavioral responses operated at magnitudes comparable to mandated restrictions; this supports the broad parameter range, not a specific functional form.

4 Identification: Debt Equation

4.1 Omission of quarter FE

Defense. The argument in the proposal — that S_k is not a regressor and y_{ik} absorbs common dynamics — is correct but incomplete. Common monetary-policy shocks affect debt directly through the interest channel rb_k . The constant r in Equation (5) implicitly assumes a uniform discount rate.

Robustness. Add country-specific time-varying r_{ik} from 10-year sovereign yields; re-estimate. If $\hat{\kappa}$ coefficients are stable, the constant- r assumption is innocuous. Alternatively, add quarter FE and verify that the cost coefficients remain identified (variation in fiscal deployment within quarter across countries).

4.2 Reverse causality on debt

Defense. Countries with rising borrowing costs may have curtailed CP loans, biasing $\hat{\kappa}^{loans}$ downward. The country FE absorbs cross-country differences in debt levels but not within-country time variation.

Test. Use the lagged sovereign-yield spread (over Bund or 10-year US Treasury) as a control. If $\hat{\kappa}^{loans}$ rises after controlling for borrowing costs, the bias is confirmed but its magnitude is bounded.

4.3 Take-up rate sensitivity

Defense. The 35% take-up is the OECD-aggregate estimate from ECB and IMF Fiscal Monitor. National variation is acknowledged.

Sensitivity table. Re-estimate $\hat{\kappa}^{guar}$ at take-up rates of {15%, 25%, 35%, 50%, 75%}. Report whether the policy ranking (guarantees cheaper than direct expenditure) survives. The structural prediction is that even at 75% take-up, guarantees remain cheaper than above-the-line CP because announcement effects preserve capacity at zero realized cost.

Country-specific take-up. Where national data exist (Italy MEF, Germany BMF), use country-specific rates as a further robustness check.

4.4 Interpretation of $\hat{\kappa}^{above} \approx 0.42$

Defense. Each euro of above-the-line CP generates only 42 cents of debt because the remaining 58 cents is offset by the output effect itself: preserved firms pay taxes, retain employees who pay social contributions, and reduce mandatory transfer outlays. The automatic stabilizer $\gamma_y = -0.21$ partially captures this offset; the residual reflects the structural-budget improvement from preserved capacity.

Decomposition check. Compute $\hat{\kappa}^{above} - \gamma_y \cdot (\partial y / \partial F^{CP,above})$. If the residual is close to zero, the offset is fully explained by the output effect; if not, additional channels (e.g., lower bankruptcy losses) remain.

4.5 Statistical separation κ^{loans} vs. κ^{guar}

Defense. The point estimates (0.352 vs. 0.095) differ by 0.257. With clustered SE of 0.083 and 0.164, the SE of the difference under independence is $\sqrt{0.083^2 + 0.164^2} = 0.184$, yielding $t \approx 1.40$, $p \approx 0.16$ two-sided. The difference is not significant at conventional levels.

Implication. The policy claim must be reformulated as: “contingent instruments *plausibly* dominate direct expenditure, conditional on the take-up adjustment.” A formal test of $H_0 : \kappa^{loans} = \kappa^{guar}$ should be reported, and the AER-style claim weakened to reflect the imprecision.

Improvement. Stack the two coefficients in a constrained-vs.-unconstrained Wald test that accounts for the covariance between $\hat{\kappa}^{loans}$ and $\hat{\kappa}^{guar}$ from the same regression. The covariance is typically negative (collinearity), reducing the SE of the difference.

5 Measurement and Data

5.1 HP filter on a five-year window

Defense. HP with $\lambda = 1600$ on quarterly 2015–2019 data (20 observations) is short. End-point bias is real, but mitigated by projecting the trend forward rather than re-estimating it on pandemic data — so the trend at the boundary is anchored on the pre-pandemic period only.

Robustness. Replicate with: (i) Hamilton (2018) regression filter; (ii) linear trend on $\log(\text{GDP})$; (iii) production-function output-gap from OECD Economic Outlook (which uses a separate methodology). Report $\hat{\eta}_p$ for each. The structural claim should be invariant to the specific output-gap construction.

5.2 Excess-mortality baseline

Defense. The Msemburi et al. (2023) WHO methodology is the most validated cross-country baseline, but national point estimates differ from OWID and EuroMOMO by up to 30%.

Robustness. Re-estimate with three baselines: WHO (current), OWID linear projection, EuroMOMO (where available, ~ 25 European countries). If $\hat{\beta}_d$ is sign-stable and within $\pm 30\%$ of the WHO-based estimate, the result is robust.

5.3 Variant-time-varying IFR for θ recovery

Defense. The IFR fell substantially from Wuhan to Delta to Omicron, and pre-vaccine Levin et al. (2020) IFRs overstate fatality from late 2021 onward. The user’s existing approach uses country-specific age-adjusted IFRs that hold the age structure fixed but not the variant.

Correction. Apply variant-share weights from GISAID and the variant-IFR estimates from IHME (2022) to produce a time-varying IFR_{ik} . Re-recover θ_{ik} . Report whether the wave-ranking of θ is preserved (memory: “correct ranking over waves” is one of the requirements for $\hat{\theta}$ as state variable).

5.4 Stringency Index measurement error

Defense. Classical measurement error on a regressor attenuates the coefficient toward zero. The reported $\hat{\alpha}_S = -0.034$ is therefore likely a lower bound on the true direct output cost of containment.

Alternative aggregations. Replicate with: (i) the eight individual sub-indices entered separately; (ii) principal-component aggregation; (iii) the Goldstein–Rivers index (which weights by domain-specific elasticities). If $\hat{\alpha}_S$ scales appropriately and $\hat{\eta}_p$ remains stable, the measurement error is non-fatal.

5.5 Classification of 2,301 measures

Defense. Single-coder classification is a known limitation of micro-policy databases. The code-book (already on GitHub) defines the classification rules, but inter-rater reliability is not yet established.

To implement. Have a research assistant (or a senior co-author) re-classify a random 10% sample. Report Cohen’s κ as inter-rater reliability. Cohen’s $\kappa > 0.7$ is the conventional threshold for substantial agreement in policy-coding studies. If $\kappa < 0.7$, identify the categories with disagreement and refine the rules.

5.6 Subnational fiscal measures

Defense. The database covers central-government measures. Subnational support (US states, German Länder, Italian regions) is not included.

Magnitude check. For a subset of countries (US, Germany, Italy, Spain), compare central-only vs. general-government fiscal series from the OECD. If the omitted subnational share is $< 10\%$ of the cumulative total, the bias is bounded. If larger, this should be acknowledged as a measurement limitation, particularly for federal systems.

5.7 Real-time vs. revised GDP

Defense. The output gap is constructed from the latest OECD vintage. Revisions to pandemic-era GDP have been substantial in some countries (Mexico, Chile).

Robustness. Use the OECD Real-Time Database for the 2020–2022 vintage and re-construct the gap. Report $\hat{\eta}_p$ on real-time vs. revised data. This addresses both the revision concern and the related concern that policy was set on real-time information.

6 Optimal Control and iLQR

6.2 Linearization point

Defense. iLQR linearizes around a nominal trajectory. The natural choice is the country-specific observed trajectory; the algorithm iterates from this initialization to convergence.

Stability check. Re-solve from three initializations: (i) the observed trajectory; (ii) zero policy ($S = F = 0$); (iii) the OECD-average trajectory. If the converged optimum is invariant to initialization, the problem is well-conditioned. If not, multiple local optima exist and should be reported.

6.3 Quadratic loss on mortality

Defense. Quadratic loss is the LQR-imposed functional form, not a structural prediction. The economic interpretation is that the planner is risk-averse over mortality outcomes, which is consistent with VSL frameworks where the welfare cost of additional deaths rises in aggregate exposure (capacity constraints, healthcare overload).

Sensitivity. Re-solve with linear loss on d (constant marginal value, standard VSL) and report whether the optimal F^{CP}/F^{DI} split shifts. The mortality channel enters through β_d in the output equation; its weight in the planner’s objective is the multiplier in front, not the linear-quadratic structure per se.

6.4 Discount factor β

Defense. Quarterly discount factor $\beta = (1/(1 + r_a))^{1/4}$ with $r_a = 0.02$ as the OECD-average pre-pandemic real rate, yielding $\beta \approx 0.995$. The interest rate r in the debt equation should be calibrated to the same long-run real rate for internal consistency.

Sensitivity. Solve at $r_a \in \{0.00, 0.02, 0.04\}$. Optimal policy is typically invariant within this range over an 8-quarter horizon; the welfare comparison is more sensitive.

6.5 Terminal cost on debt only

Defense. The terminal weight $Q_N = \beta^N \text{diag}(0, 0, W_b, 0, 0)$ encodes that debt persists at $(1 + r)$ while output gaps and mortality decay endogenously after vaccination. This captures the central economic feature of the post-pandemic state.

Concession. Output-gap persistence beyond N is omitted. If the persistence channel η_p remains active in the recovery phase (no reason to think otherwise), some terminal weight on y_N is justified.

Robustness. Add $w_y^N \cdot y_N^2$ at calibrated weight; verify that the optimal trajectory is qualitatively unchanged. If sensitive, this should be modeled rather than absorbed.

6.6 Existence and uniqueness of the optimum

Defense. The problem is non-convex due to the bilinear interactions $S_k y_k$ and $F_{k-1}^{CP} y_k$. iLQR is a local-search algorithm and may converge to local minima.

Multi-start verification. Solve from ≥ 10 random initializations within the feasible set; verify that the converged optimum is the same. Report the dispersion of converged costs.

Alternative algorithm. Cross-validate with a global optimizer (e.g., genetic algorithm or simulated annealing) on a coarse grid. If the iLQR optimum is at or below the global-optimizer cost, the local-search result is at least as good.

7 Calibration and Welfare Decomposition

7.1 Revealed-preference circularity

Defense. The OECD-average preference weights are not normative. They are a positive description of revealed average behavior. Country deviations are interpreted relative to the OECD-average behavior, not relative to a normative ideal.

Two-step decomposition. (i) Country deviations from OECD-average are attributable to either differing preferences (position along frontier) or efficiency losses (distance from frontier). (ii) The OECD-average itself can then be benchmarked against an external normative anchor: e.g., a planner with VSL-based mortality weights, or a planner minimizing only debt subject to mortality and output constraints. The second benchmark addresses the circularity.

Reframing. State explicitly in the paper that preference weights at the OECD average are positive, not normative; the welfare interpretation requires the second-step external benchmark.

7.2 Identifiability of preference vs. efficiency

Defense. The decomposition requires the country-specific frontier to be uniquely traceable. iLQR with continuous preference weights traces the frontier as a one-parameter (or two-parameter, depending on dimensions) family. Two distinct preference vectors $w \neq w'$ can produce frontiers that intersect at a given realized point, but these correspond to different objective values: only one minimizes total cost at the realized point.

Identifying restriction. The decomposition assigns each realized point to its closest frontier point in the cost-weighted Mahalanobis metric; the residual distance is the efficiency loss. This is well-defined as long as the frontier is convex (which it is, by LQ structure under the convex-hull constraint).

7.3 Welfare units

Defense. The relative weights determine the policy ranking. Without external anchors, the welfare comparison is internal. To provide an external anchor: convert excess mortality to monetary units using country-specific VSL (e.g., VSL $\approx$ 9M USD for the US, scaled by per-capita income for other countries); convert debt to monetary units using a deadweight-loss-of-taxation parameter (e.g., 0.20 per dollar raised, following Saez–Stantcheva).

Reporting. Report the welfare comparison in two units: (i) model-internal (preference-weighted), (ii) externally-anchored (VSL + DWL). If qualitative rankings agree, the result is robust to the choice of units.

7.4 Heterogeneity in preferences vs. parameters

Defense. The Mundlak decomposition shows between-country variation in fiscal deployment is correlated with crisis severity (sign-flips on $\hat{\eta}_p$ in between estimates). This is parameter heterogeneity, not preference heterogeneity.

Correction. Allow country-specific $\eta_{p,i}$ via random coefficients (Swamy 1970) or via mixed-effects estimation (Wooldridge–Pesaran approaches). Compare the dispersion in $\hat{\eta}_{p,i}$ to the dispersion in implied preferences. If parameter heterogeneity dominates, the preference–efficiency decomposition is biased toward attributing parameter differences to preferences.

8 Cross-Country Heterogeneity and External Validity

8.1 Pooled estimation across 38 OECD countries

Defense. Pooling buys statistical power; heterogeneity costs interpretability. The within-country, within-quarter identification minimizes the role of cross-country differences in levels but not in slopes.

Slope-heterogeneity tests. (i) Pesaran–Yamagata (2008) test for slope heterogeneity. (ii) Country-by-country estimation of $\hat{\eta}_p$ where T permits (impossible at $T = 10$, but feasible for the level parameter $\hat{\alpha}_S$). (iii) Re-estimate on subsamples: G7, EU, non-EU OECD. Report whether the headline result holds within subsamples.

8.2 Sectoral composition

Defense. Sectoral mix is absorbed by the country fixed effect to the extent that it is time-invariant. Time variation in sectoral composition (e.g., service-sector decline during lockdowns)

is not absorbed.

Control. Add the share of contact-intensive services (NACE I, R, S) from Eurostat as a time-varying covariate. If $\hat{\eta}_p$ is robust, sectoral composition is innocuous.

8.3 Monetary policy

Defense. Monetary policy enters the debt equation through the interest rate r . In the output equation, monetary effects on y are absorbed by quarter fixed effects to the extent that they are common (the major QE programs were synchronized: Fed, ECB, BoE, BoJ, SNB). Country-specific monetary divergence (e.g., emerging-market OECD members tightening) is not absorbed.

Control. Add the country-specific real policy rate as a time-varying covariate. Subset to advanced-economy OECD (excluding Mexico, Chile, Turkey, Colombia) as robustness.

8.4 Labor-market institutions

Defense. Heterogeneous EPL implies the persistence channel of CP varies in strength across countries: high-EPL economies have less to preserve at the margin (matches are already protected). The pooled $\hat{\eta}_p$ is a weighted average.

Test. Interact $F_{i,k-2}^{CP}y_{i,k-1}$ with the OECD EPL Index (pre-pandemic value). A negative interaction (CP more effective in low-EPL economies) supports the structural interpretation. The result should be reported as a structural prediction validated, not a fragility.

9 Robustness and Inference

9.1 Cluster count

Defense. 38 clusters is at the lower edge of standard cluster-robust asymptotics. CRV1 critical values are oversized; CRV3 (Bell–McCaffrey 2002) and wild cluster bootstrap (Cameron–Gelbach–Miller 2008) provide better inference.

Plan to report. (i) CRV1 (current), (ii) CRV3 via the `summcust` package, (iii) wild cluster bootstrap with Webb weights and 999 replications, (iv) Ibragimov–Müller (2010, 2016) randomization inference using country-pair permutations. The headline claim should be that $\hat{\eta}_p$ remains significant under all four. Memory: the user’s R environment already has `summcust` for CRV3.

9.2 DCDH in continuous-treatment TWFE

Defense. The de Chaisemartin–D’Haultfœuille (2020) negative-weight critique was originally for binary staggered treatment. Extensions exist for continuous treatments (de Chaisemartin–D’Haultfœuille–Vazquez-Bare 2024). The relevant diagnostic is whether the TWFE estimator places negative weights on individual treatment effects.

Implementation. Compute the weights using the `TwoWayFEWeights` package (R) or `twowayfeweights` (Stata). Report the share of negative weights and their magnitude. If small, the TWFE-as-weighted-average interpretation holds.

Alternative estimator. The de Chaisemartin–D’Haultfœuille (2024) DID-multiple estimator handles continuous treatment without the weighting problem. Report as robustness.

9.3 LP vs. TWFE

Defense. Local projections (Jordà 2005) estimate impulse responses non-parametrically and are robust to dynamic mis-specification. With $T = 9$ usable quarters, LP at horizon h requires $T - h$ observations per unit; horizons $h \geq 4$ are imprecise.

Plan. Report LP at $h \in \{0, 1, 2\}$ for the level coefficients $\alpha_S, \alpha_{DI}, \beta_d$. The persistence interaction is harder to identify in LP; document this and treat TWFE as the main specification, LP as robustness for level parameters.

9.4 System GMM

Defense. System GMM is the canonical dynamic-panel estimator for this setting. The Roodman (2009) instrument-count concern requires lag truncation.

Specification. Lags 2–3 of the endogenous variables as GMM-style instruments, country dummies as standard instruments. Collapsed instrument matrix. Target instrument count ≤ 30 to stay below $N = 38$. Report the Hansen J , the AR(1) and AR(2) tests.

Caveat. If the AR(2) test rejects, GMM identification fails and the result should not be invoked. The Nickell-corrected LSDV becomes the preferred robustness in that case.

9.5 Outliers

Defense. Greece, Italy, and Spain experienced extreme output gaps; Mexico had unusual fiscal composition (low overall, predominantly DI). These observations have high leverage.

Plan. Leave-one-country-out estimation; report the distribution of $\hat{\eta}_p$ across the 38 leave-one-out estimates. Report the maximum and minimum, with the corresponding country. Memory mentions “18-item robustness checklist”: this is part of it.

10 Connection to Existing Literature

10.1 Guerrieri et al. (2022)

Defense. The reduced-form estimate $\hat{\alpha}_{DI} \approx 0$ is consistent with their prediction that DI fails when supply is constrained. They also show that DI can stabilize incomplete-markets economies through insurance against income shocks. The reduced form does not separate the two channels, but the policy implication is the same: DI is the inferior instrument when capacity is the binding margin.

Reference. Cite the supply-constrained mechanism as the primary theoretical source; acknowledge the insurance channel as complementary.

10.2 Chetty et al. (2020)

Defense. Their real-time evidence on US direct transfers showed limited spending response during containment. The TWFE estimate is qualitatively consistent: $\hat{\alpha}_{DI} \approx 0.18$ (insignificant) over the pandemic window.

Quantitative comparison. Their MPC estimates from the CARES checks were ~ 0.40 for high-income, ~ 0.10 for low-income. The implied aggregate spending multiplier was ~ 0.3 in their sample, also far below pre-pandemic estimates. The order of magnitude matches.

10.3 Faria-e-Castro (2021)

Defense. Their non-linear DSGE shows that pandemic-era fiscal multiplier rankings differ from normal times: liquidity assistance (CP-equivalent) dominates transfers (DI) for employment stabilization. The reduced-form result confirms this prediction at the OECD scale and on output rather than employment.

New contribution. Their result is structural and US-specific; this paper provides cross-country reduced-form evidence. The complementarity is the contribution.

10.4 Deb et al. (2021)

Defense. They classify by IMF budgetary category (above-the-line vs. below-the-line); this paper classifies by economic transmission mechanism. The two classifications cut across each other: above-the-line CP and above-the-line DI are different in this paper but identical in theirs.

Reconciliation. Re-run the headline specification with their classification (above-the-line, below-the-line) and report. If their headline result (emergency-lifeline above income-support during strict containment) holds in this sample, the contribution is the refined classification. If not, the contribution is the disagreement.

10.5 Ramey–Zubairy (2018)

Defense. Their multiplier of 0.6–0.8 is for normal-times government spending, identified from US historical data using military news shocks. The pandemic-era $\hat{\alpha}_{DI} \approx 0.18$ is far below.

Reconciliation. The discrepancy is structural: pandemic DI flows into a contact-restricted economy where the consumption margin is bound. This is precisely the Guerrieri et al. (2022) prediction. The result is therefore a successful out-of-sample test of the supply-constrained mechanism.

11 Paper 2: The Wrong Side of Disruption

11.1 Identification (Digital Dividend IV)

Q1.1: Exclusion restriction. Pre-treatment terrestrial-TV share correlates with rurality, age, income. Defense: the IV is the *interaction* between national switch-off timing and pre-treatment share, not the share itself. The share enters as a fixed effect; identifying variation comes from the differential timing of capacity gain across cities with the same pre-treatment share. Robustness: control for time-varying urbanization, demographic structure, and income from Eurostat NUTS2 data.

Q1.2: Shift-share identification. Goldsmith-Pinkham et al. (2020) requires exogenous shares; Borusyak–Hull–Jaravel (2022) requires exogenous shocks. The cleaner case here is BHJ: the national switch-off dates are determined by EU-level spectrum-allocation policy, plausibly exogenous to local drug markets. The effective number of independent shocks is the number of national switch-off dates (~ 27 for the 800 MHz dividend, ~ 27 for 700 MHz), which is small.

Q1.3: First-stage strength. Plan to report (i) the standard first-stage F -statistic, (ii) the Olea–Pflueger (2013) effective F , (iii) the Lee–McCrary–Moreira–Porter (2022) tF threshold for valid 5% inference. If the effective $F < 23.1$, report tF-adjusted critical values.

Q1.4: Colombian supply confounding. The 2014–2020 Colombian production expansion is a global supply shift; cross-city variation in cocaine purity within the same country and year

cannot be driven by it. The IV identifies city-level capacity gains, which generate within-country variation orthogonal to global supply trends.

Q1.5: Mechanism specificity. The Digital Dividend instruments mobile internet capacity, not platform use. The LATE is the effect of mobile internet penetration on consumption, not the platform mechanism per se. The platform-mechanism claim requires the complementary food-delivery DiD and event-study designs. Acknowledge the LATE explicitly.

11.2 Data and Outcomes

Q2.1: SCORE limitations. Single-week annual campaigns yield ~ 50 observations per week per city annually, 7 days per year. Daily seasonal variation is large; annual aggregation is the only feasible analytical unit. For high-frequency event studies, the daily-monitoring subsample (Switzerland, UK) is the binding sample.

Q2.2: Selection into SCORE. Cities self-select. Compare the SCORE city distribution to Eurostat NUTS2 demographics; weight observations by population and country if necessary. Acknowledge as a sample-representativeness limitation.

Q2.3: COVID-19 contamination of the second window. The 2020–2021 lockdowns affected both physical and digital drug markets. The IV identifies the long-difference effect of capacity gain, which does not depend on the contemporaneous lockdown intensity. Report the IV estimate excluding 2020–2021 quarterly data as robustness.

Q2.4: Substance heterogeneity. Cocaine, MDMA, methamphetamine, cannabis differ in market structure and platform reliance. Plan: estimate the IV separately by substance. The platform mechanism predicts strongest effects for high-purity, contact-discreet substances (cocaine, MDMA) and weakest for cannabis (already non-contact via established channels).

Q2.5: Spillovers. A buyer in city A may use a Telegram dealer in city B. SUTVA is violated. Magnitude check: the effect should attenuate at finer geographic resolution if spillovers dominate. Report city-level vs. NUTS2-level vs. country-level estimates.

11.3 Complementary Designs

Q3.1: Food-delivery launch DiD. Cities with growing young populations attract food delivery and recreational drug use. Control for time-varying demographic structure (Eurostat). Use only the launch sequence within the 2-year window before and after each launch (event-time normalization), not the levels.

Q3.2: Durov-arrest event study. Daily wastewater monitoring exists in ~ 10 European cities. The effective sample for the high-frequency event study is therefore $10 \text{ cities} \times \sim 60 \text{ days} = 600$ observations around the event. Statistical power is limited; report as suggestive complement, not main evidence.

12 Big-Picture Defense Questions

12.1 Why this paper, why now?

Defense. Three policy margins remain active: (i) the fiscal-composition design problem for the next pandemic or large-scale supply shock; (ii) the choice between direct expenditure and contingent liabilities for crisis response in a high-debt environment; (iii) the structural-budget legacy of the pandemic itself. The first informs preparedness; the second informs current debt-sustainability analysis; the third is unresolved in the existing literature.

12.2 Internal vs. external validity

Defense. The within-country, within-quarter identification recovers structural parameters that are then deployed in the cross-country optimization. The cross-country claim — that contingent instruments dominate direct expenditure — is a structural prediction conditional on the estimated parameters being country-invariant. The Mundlak decomposition documents that this is approximately true (within-country sign of $\hat{\eta}_p$ stable; between-country sign-flips reflect crisis severity, not parameter heterogeneity).

Concession. If parameter heterogeneity is large (Section 7.4 robustness), the cross-country policy ranking is weakened. Report the ranking with country-specific parameters as a robustness check.

12.3 What would change your mind?

Defense. Three falsifiers: (i) $\hat{\eta}_p \rightarrow 0$ under any of the bias corrections in Section 3.1 (Nickell, AB, BB, Kiviet); (ii) $\hat{\kappa}^{guar}$ rises above $\hat{\kappa}^{above}$ under alternative take-up rates within ± 25 percentage points of the central estimate; (iii) the asymmetric persistence test (Q3.7) shows the effect is driven entirely by the recession regime in a way that does not generalize. Each is testable; the structural claim survives if and only if it survives all three.